Study Design Document: Study I

Task Switching comfort using colored light

Purpose of study (research hypothesis/questions)

The purpose of this study is to investigate how color-coded background light can influence user experience and comfort during cognitive task switching.

Research questions are formulated as follows:

I. How can cognitive task switching be supported by providing environments differentiated by color?

II. Do differently colored environments lead to a more enjoyable work setting in multitasking scenarios?

III. How does the comfortability of users change with introducing color-based work environments?

The research hypotheses are structured as follows:

I. Participants will report lower levels of perceived workload under coloured lighting compared to the uniform white lighting.

II. Participants will report higher comfort levels under coloured lighting compared to the uniform white lighting.

III. Coloured lighting will improve task performance compared to the uniform white lighting.

Participant profile

Participants are selected based on these criteria:

- **Inclusion criteria**
  - Participants must have basic PC and keyboard experience, sufficient English language skills to read and type a text, and be older than 18 years old.
  - Participants must have normal or
- **Exclusion criteria**
  - Participants who self-report neurological or psychiatric conditions that significantly impair concentration (e.g., schizophrenia, ADHD, etc.) will be excluded.
  - Participants who self-report a history of photosensitive epilepsy or seizures triggered by visual stimuli will be excluded.

    - Participants who self-report color blindness or significant visual impairments that are not corrected.

Before entering the study potential participants will be provided with the following information via an information sheet/the online sign up site:

This study involves working at a computer with colored LED lighting. The light may change color occasionally but will not strobe rapidly.

If you are a person with color blindness, have a history of photosensitive epilepsy, seizures triggered by visual stimuli, or any condition that may be aggravated by changing lights, you are advised not to participate in this study.

Participation is voluntary and can be discontinued any time before or during a study (participants will also be informed of this through an informed consent form), all collected data will be stored anonymously and used only for this study.

Compensation plan

- We participate in other students' studies.
- Additionally, light refreshments (e.g. sweets, etc.) will be provided. No monetary compensation will be offered.

Methodology

Study Design

Quantitative	**Formative** and **Summative**

Experiments

Within-subject	**Randomized order**

Estimated number of participants	**≥ 20**

We use a within-subject experiment with one IV (light condition), where each participant completes 2 blocks of tasks: one under coloured light and one under baseline light. The order of conditions is counterbalanced and “randomly” assigned across participants to control learning effect, fatigue and general order effect. In each block one main task and 2 secondary tasks are performed to simulate cognitive task switching.

Variables

Independent variables	Dependent variables	Confounding variables
Light condition: (baseline: white light for all task switching; coloured light: main task with white light, secondary tasks with blue/green)	**Typing speed:** continuous, amount of the correctly transcribed characters in the main task (correct characters per minute) **Number of typing errors:** discrete, typing errors in main task **NASA Task Load Index**: for comfortability **Subjective comfort and flow:** assesses state flow	**Individual task switching ability:** some participants may feel task switching easier than others **Rest/break time:** fatigue can affect performance if one condition is always experienced at the same position **Pre-existing mental/neurological disorders:** Schizophrenia, ADHD, Alzheimer, colour blindness, visual impairments, etc.

Material & Apparatus

Technical setup

- Laptop
- LED light with multiple colors and (remote) control
- Desk & chair
- Experiment Suite Software

Task Setup

- Tasks to be performed by users

- **Main task:** Transcribe “The Adventure of Three Students” by Doyle
- **Smoke Task B:** Identify consonant/vowel vs. odd/even
- **Smoke Task C:** Spot-the-difference

Color Setup

Task	Color	Light Bulb Setting
Main	White	Brightest
Smoke Task A	Blue	Brightest
Smoke Task B	Green	Brightest

Questionnaires

Please use this link to view the Questionnaire setup: https://docs.google.com/forms/d/18tITw-ERHPjfV3DBSdQG3BmAOOPjKd9vg0Q-j9_AlGI/edit

Demographic questionnaire

- Age
- Gender
- Profession
- Highest education level
- Medical condition (ask for all mentioned as exclusion criteria)

The **NASA task load index** (NASA TLX) is a tool for measuring and conducting a subjective mental workload (MWL) assessment. It enables the determination of a participant's MWL while they are working on a task. In order to determine an overall workload rating, it evaluates performance on six dimensions. https://digital.ahrq.gov/health-it-tools-and-resources/evaluation-resources/workflow-assessment-health-it-toolkit/all-workflow-tools/nasa-task-load-index

- Mental Demand – How much mental and perceptual activity was required (thinking, deciding, calculating, remembering, looking, searching, etc.)?
- Physical Demand – How much physical activity was required (pushing, pulling, typing, controlling, activating)?
- Temporal Demand – How much time pressure did you feel due to the rate or pace of the task elements?

- Performance – How successful do you think you were in accomplishing the goals of the task? (High = perfect performance; Low = failure.)
- Effort – How hard did you have to work (mentally and physically) to achieve your level of performance?
- Frustration – How insecure, discouraged, irritated, stressed, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel?
    - <u>Reformulated Questions:</u>
    - Mental Demand: “How mentally demanding was the task?”
      1 = low, 20 = high
    - Physical Demand: “How physically demanding was the task?”
      1 = low, 20 = high
    - Temporal Demand: “How hurried or rushed was the pace of the task?”
      1 = low, 20 = high
    - Performance: “How successful were you in accomplishing what you were asked to do?”
      1 = very unsuccessful, 20 = very successful
    - Effort: “How hard did you have to work to accomplish your level of performance?”
      1 = not hard at all, 20 = very hard
    - Frustration: “How insecure, discouraged, irritated, or stressed did you feel during the task?”
      1 = not at all, 20 = very

Flow and task-related well-being will be measured with the **Flow Short Scale**, which assesses state flow using 7-point Likert items. The FSS consists of a total of 16 items. Items 1-10 assess components of the flow experience, in addition to the flow assessment itself, items 11-13 measure a Worry component and items 14-16 are intended to measure the balance of skill and demands.

https://www.testarchiv.eu/en/test/9004690

Flow-Items (7-point scale, from "strongly disagree" to "strongly agree"):

1. I feel just the right amount of challenge.
2. My thoughts/activities run fluidly and smoothly.
3. I don’t notice time passing.
4. I have no difficulty concentrating.

5. My mind is completely clear.
6. I am totally absorbed in what I am doing.
7. The right thoughts/movements occur of their own accord.
8. I know what I have to do each step of the way.
9. I feel that I have everything under control.
10. I am completely lost in thought.

Worry-Items (7-point scale, from "strongly disagree" to "strongly agree")

11. Something important to me is at stake here.
12. I won't make any mistakes here.
13. I am worried about failing.

Balance of skill and demands (9-point scale)

14. Compared to all other activities which I partake in, this one is … ("easy" to "difficult").
15. I think that my competence in this area is … ("low" to "high").
16. For me personally, the current demands are … ("too little" to "too high").

Structure & Timeline

Preparation required before the study starts:

- Informed consent form
- Demographic data
- Assessment of influential factors e.g. experience/expertise, drinking/smoking behavior

Timeline of each task

Task	Subject	Time
Main	Initial familiarity period. Participants get comfortable with the main task.	2min
B	Participants are asked to switch to smoke task B.	30sec
Main	Participants are asked to switch back to the main task.	30sec

Task	Subject	Time
A	Participants are asked to switch to smoke task A.	30sec
Main	Participants are asked to switch back to the main task.	30sec
B	Participants are asked to switch to smoke task B.	30sec
Main	Participants are asked to switch back to the main task.	30sec
A	Participants are asked to switch to smoke task A.	30sec
Main	Participants are asked to switch back to the main task.	30sec
Q	Questionnaire + data saving	2min
	Total	8min

Timeline of experiment

- Participants get introduced to the study
- Participants familiarize themselves with the task
- Break
- Participants complete the first iteration of the tasks and complete a questionnaire
- Break
- Participants complete the second iteration of the tasks and complete a questionnaire

Phase	Subject	Material	Time
Prep	Participants will be informed about the experiment and its structure and will sign a consent form (informed consent form produced by the Frankfurt University of Applied Sciences - https://hci-studies.org/informed-consent-generator/) Collection of demographic data and “influential factors”.	Consent form	2min
	Explanation round • Explain main task • Explain smoke tasks		3min
	Let participants figure out the mechanics and ways of interaction of the tasks.		3min (1min each)
	Break		3min

Phase	Subject	Material	Time
Iteration 1	Task iteration 1. Randomly assigned with-light or without-light group.	Light setup Laptop setup Questionnaire	8min
	Break		3min
Iteration 2	Task iteration 2. Inverse group assignment.	Light setup Laptop setup Questionnaire	8min

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Analysis

NASA-TLX produces six subscale scores and an overall workload score. This study uses the weighted TLX (1–20 ratings plus pairwise comparisons), and an overall workload score will be computed for each lighting condition.

Flow will be measured with the **Flow Short Scale**, for each condition we will calculate a composite Flow Experience score as the mean of the designated flow items, so that higher values indicate stronger flow.

Differences between the coloured-light and baseline-light conditions in NASA-TLX overall workload, Flow Experience, and additional Likert-scale ratings (comfort, task-switching ease, task enjoyment) will be analysed using paired-samples t-tests. If the data is not normally distributed, non-parametric alternatives (e.g., Wilcoxon signed-rank tests) will be used. Where applicable, reliability and effect size measures may also be reported.

Personnel Involved & Responsibilities

For better time efficiency, everyone has two roles and every role is included twice so that we can run the study in parallel. Every team member is vetted on how the study is conducted and how the Software works and is thus able to conduct it in full.

Celina	Conduct Study - Receptionist (consent form provider, prepare participants, etc.) - Light operator
Kseniia	Conduct Study - Receptionist (consent form provider, prepare participants, etc.)

	- Light operator
Sepehr	Conduct Study - Software guru - Backup software implementation
Omar	Conduct Study - Receptionist (consent form provider, prepare participants, etc.) - Light operator
Johannes	Conduct Study - Software guru - Backup software implementation